

Assessment for Delta Operations on Salmonids

Last updated: *February 02, 2026*

For more detailed data on salmonid conditions in the Delta see corresponding webpage on [SacPAS](#).

Executive Summary

- Entrainment management season is active.
- Season Loss: 0 (0.00% of annual loss threshold) natural winter-run, 0 (0.00% of annual loss threshold) hatchery winter-run, 46.63 natural steelhead, 302.61 (5.12% of annual loss threshold) hatchery steelhead, and 1018.77 (46.34% of annual loss threshold) spring-run surrogates.
- Single-year Incidental Take Limit (ITL) Status: 0 (0.00% of 5,922 ITL) natural winter-run; 0 (0.00% of 1,301 ITL) hatchery winter-run; 46.63 (0.88% of 5,294 ITL) natural steelhead.
- Spring-run surrogate yearlings (0.5% ITL per experimental release group): Group 1: 0 (0% of 376 ITL); Group 2: 257.3 (84.64% of 304 ITL); Group 3: 2.9 (1.01% of 286 ITL).
- LAD winter-run presence in the Delta is high based on historical monitoring.
- Steelhead presence in the Delta is increasing based on historical monitoring.

Natural Winter-run Chinook

Juvenile Production Estimate

The Juvenile Production Estimate for winter-run is 1,057,452 for the current water year.

Current Status

Delta Entry Timing - Historically, as of Feb 01, 46% of length-at-date (LAD) winter-run have entered the Delta based on Knights Landing RST catch, 3% have exited the Delta based on Chipps Island Trawl Catch, and 7% of DNA confirmed winter-run have been salvaged.

Table 1: Average percent of annual emigrating population for unclipped LAD winter-run captured at monitoring locations and salvaged at Delta facilities for the past 10 years.

Species	Red Bluff Diversion Dam	Tisdale RST	Knights Landing RST	Sac Trawl (Sher- wood)	Chipp's Island Trawl	Salvage
Chinook, LAD Winter- run, Unclipped	99%	89%	86%	46%	3%	24%
Chinook, DNA Winter- run, Unclipped (Water Year)						7%

Red Bluff Diversion Dam Passage Estimate - As of Jan 28 estimated passage to date of LAD winter-run at Red Bluff Diversion is approximately 4.17 million fish. *Note that outmigration timing overlaps with spring-run migrating fish, and true winter-run abundance likely differs from these estimates.*

Delta Monitoring - Total catch of LAD winter-run at RSTs at Delta Entry (Tisdale, Knights Landing, Lower Sacramento River) between Jan 19 and Jan 29 is 13 individuals. Total catch at Sacramento Trawl and Beach Seines in the delta between Jan 20 and Jan 30 is 2 individuals. Total catch at Delta Exit at Chipp's Island between Jan 20 and Jan 30 is 0 individuals.

Annual Loss

The annual loss threshold for natural winter-run is 1% of the JPE or 10,575 fish. The single-year incidental take limit (ITL) is 0.56% of the JPE (5,922 fish) or 0.36% on a 3-year rolling average (BiOp Table 184). As of February 01, cumulative loss of genetically confirmed winter-run is 0 or 0.00% of the annual loss threshold. Cumulative loss in the past 7 days has been 0.

STARS

The Delta STARS Model is an individual-based simulation model that predicts survival, travel time, and routing of juvenile salmon migrating through the Sacramento–San Joaquin River

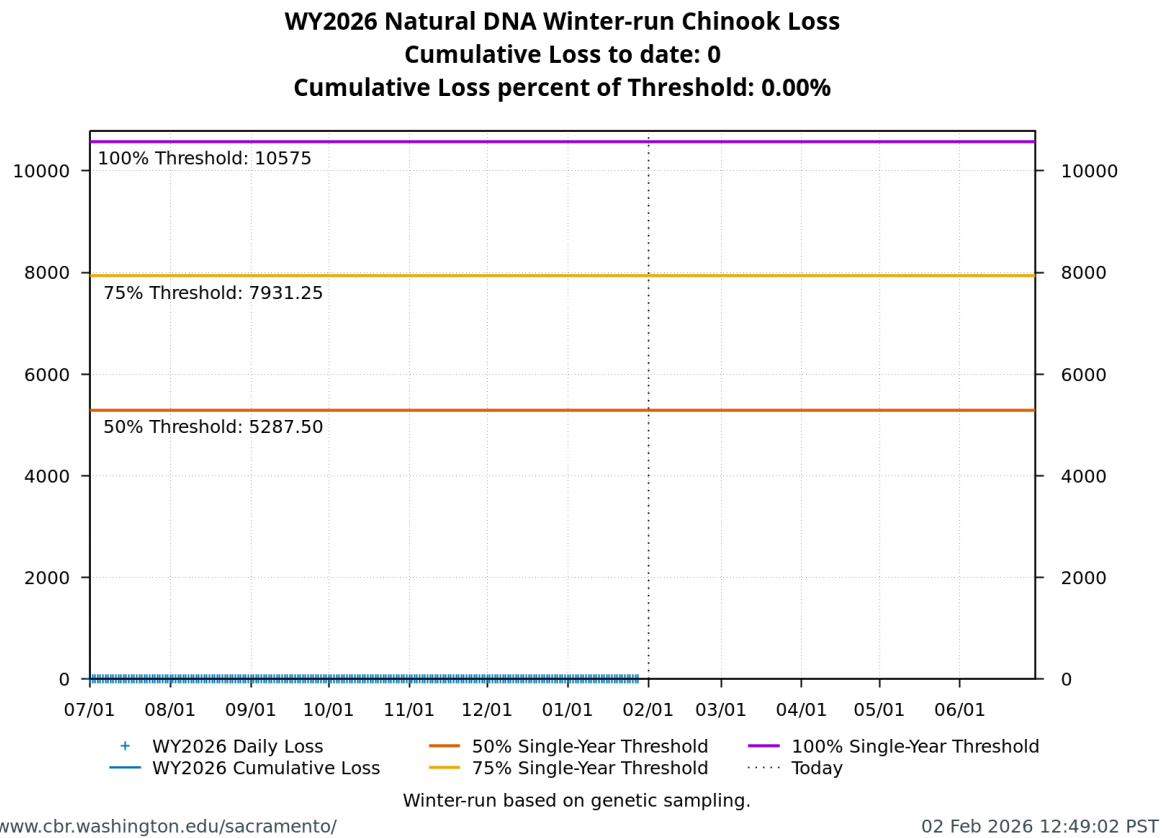


Figure 1: Cumulative loss of natural-origin winter-run for WY 2026. Cumulative loss is based on genetically confirmed winter-run captured in salvage or length-at-date winter-run in which genetic confirmation was unable to be obtained.

Delta. This model gives insight into survival and routing patterns of winter-run based on most current conditions.

As of February 01, overall through delta STARS estimated survival probability (with 80% credible intervals) is 0.6 (0.54-0.66) placing it in the 71st percentile of historical STARS survival estimates for the month of February (WYs 2018-2025). STARS estimated routing and survival probabilities (with 80% credible intervals) into the interior delta are 0.12 (0.1-0.14) and 0.32 (0.19-0.46), respectively, corresponding to the 38th and 62nd percentiles of historical February estimates (WYs 2018-2025).

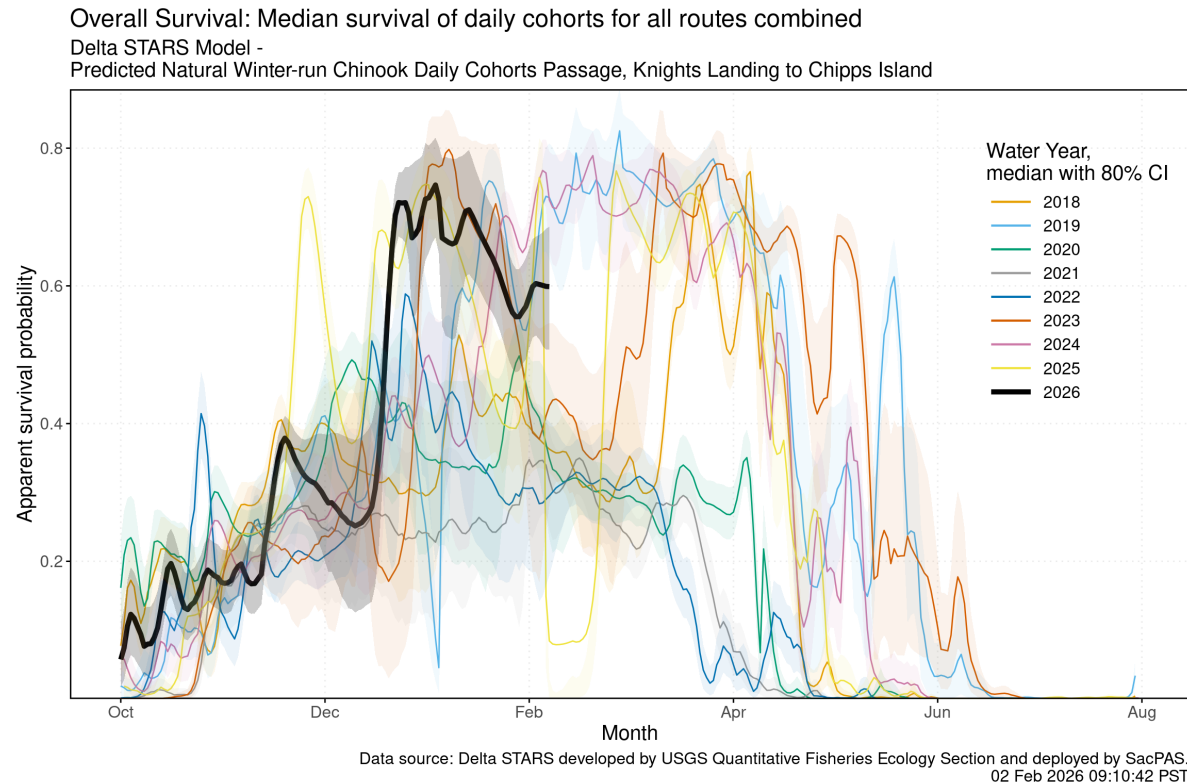


Figure 2: Estimated overall winter-run survival from Knights Landing to Chipps Island. Black line indicates the current water-year, and other colored lines correspond to past water years.

Hatchery Winter-run Chinook

Hatchery Releases

To date, no winter-run Livingston Stone hatchery releases have occurred in WY 2026.

Interior Delta Route-specific Survival Probability: Median survival of daily cohorts using the Interior Delta STARS Model - Predicted Natural Winter-run Chinook Daily Cohorts Passage, Knights Landing to Chipps Island

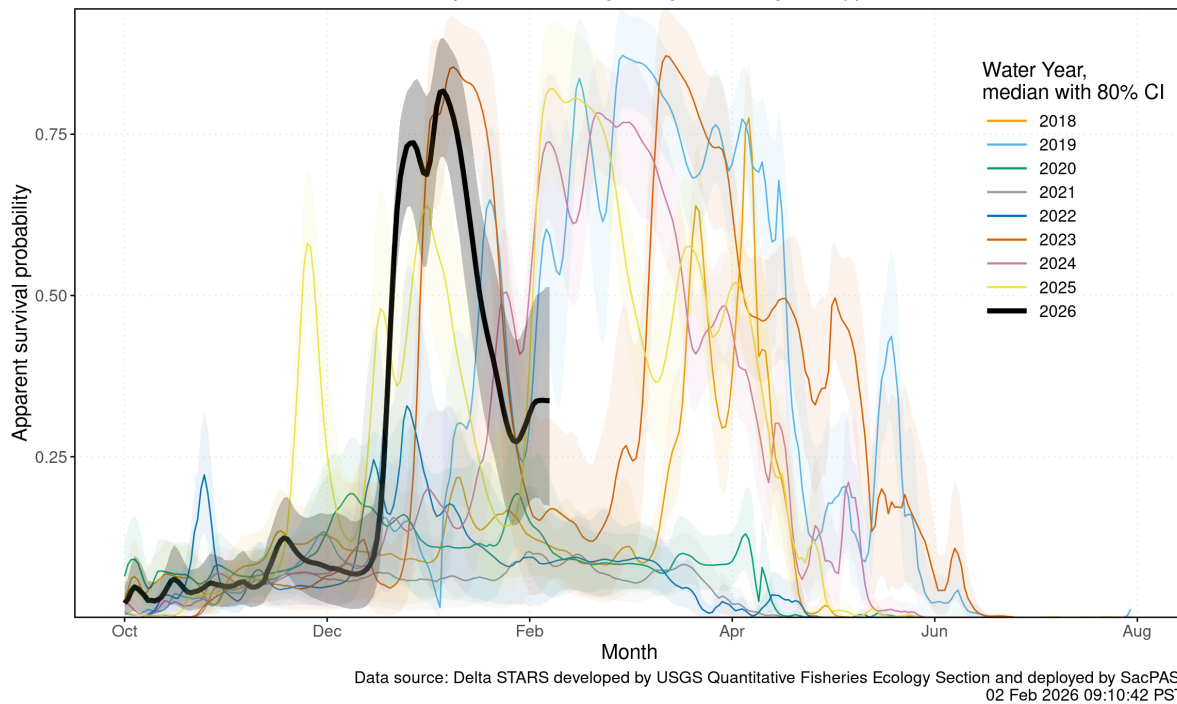


Figure 3: Estimated survival from Knights Landing to Chipps Island of simulated winter-run cohorts that route through the interior Delta. Black line indicates the current water-year, and other colored lines correspond to past water years.

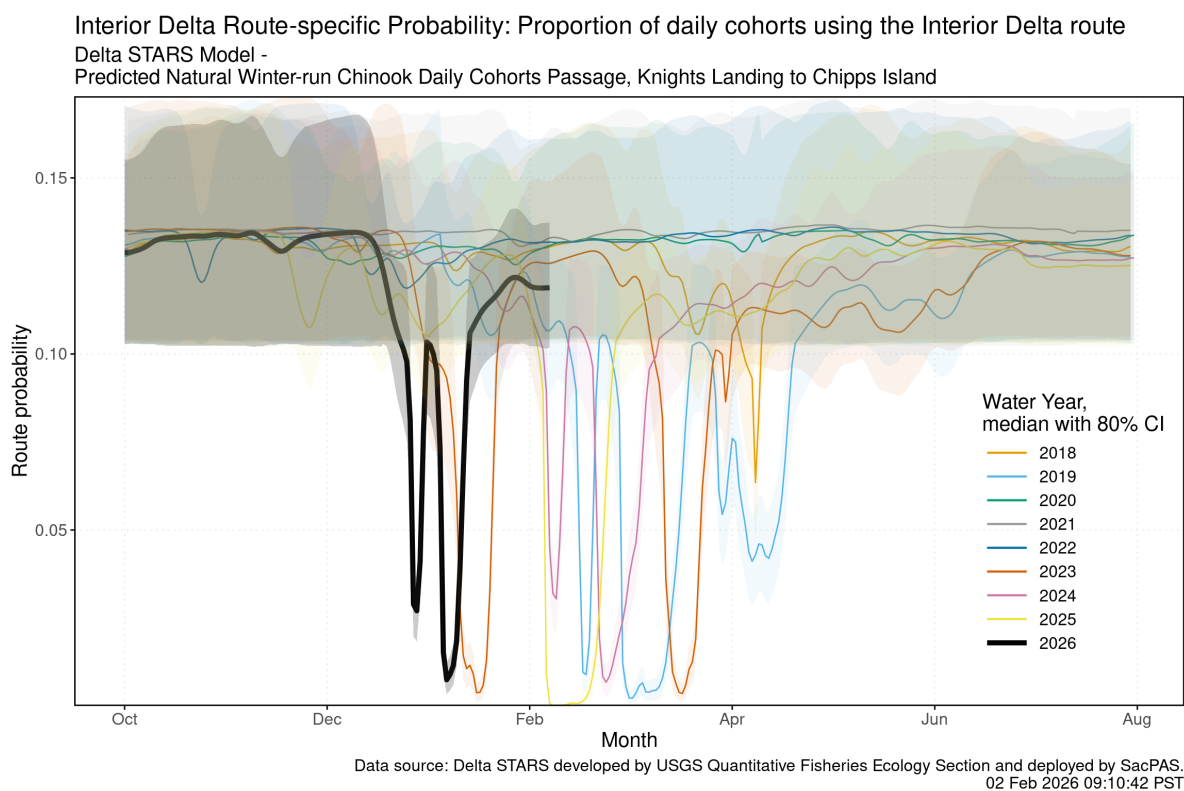


Figure 4: Estimated probability of winter-run routing into the interior Delta. Black line indicates the current water-year, and other colored lines correspond to past water years.

Juvenile Production Estimate

The Juvenile Production Estimate for hatchery winter-run is 130,096 based on current Livingston Stone production estimates. The annual loss threshold is 1% of the JPE (1,301 fish), which is the same as the single-year ITL (BiOp Table 184). Note: Physical releases have not yet occurred in WY 2026.

Annual Loss

To date, no loss has occurred as no hatchery winter-run have been released into the Sacramento River.

Natural-origin Central Valley Steelhead

Current Status

Delta Entry Timing - Historically, as of Feb 01, 16% of CCV steelhead have entered the Delta based on Knights Landing RST catch, 6% have exited the Delta based on Chipps Island Trawl Catch, and 10% have been salvaged.

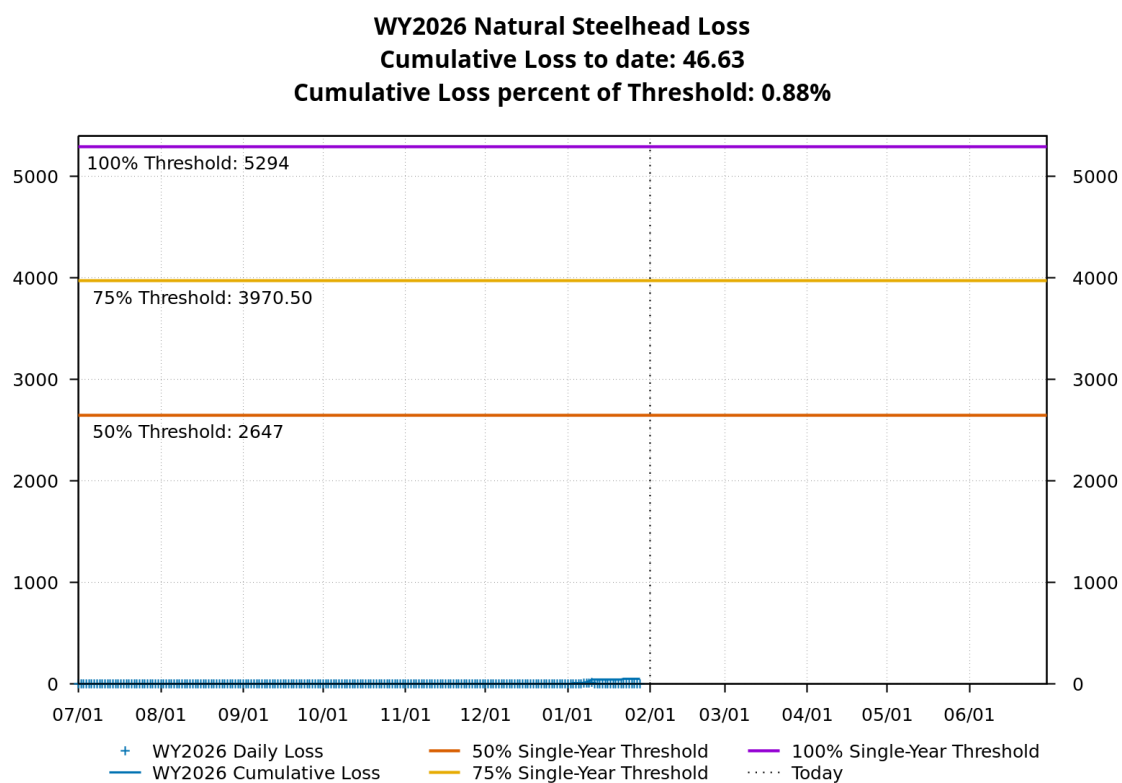
Table 2: Average percent of annual emigrating population for unclipped CCV steelhead captured at monitoring locations and salvaged at Delta facilities for the past 10 years.

Species	Chipps Island Trawl	Knights Landing RST	Red Bluff Diversion Dam	Sac Trawl (Sherwood)	Salvage	Tisdale RST
Steelhead, Unclipped	6%	33%	1%	16%	10%	28%

Delta Monitoring - Total catch of unclipped steelhead at RSTs at Delta Entry (Tisdale, Knights Landing, Lower Sacramento River) between Jan 19 and Jan 29 is 2 individuals. Total catch at Sacramento Trawl and Beach Seines in the delta between Jan 20 and Jan 30 is 0 individuals. Total catch at Delta Exit at Chipps Island between Jan 20 and Jan 30 is 0 individuals.

Annual Loss

As of February 01, cumulative loss of unclipped steelhead is 47 or 0.88% of the single-year incidental take limit (ITL). There is no annual loss threshold for natural steelhead. The single-year ITL is 5,294 juveniles or 2,319 juveniles as a 3-year rolling average (BiOp Table 184). Cumulative loss in the past 7 days has been 89.



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Figure 5: Cumulative loss of natural-origin steelhead for WY 2026. The 5,294 line represents the single-year incidental take limit (ITL), not a loss threshold.

Hatchery-origin Central Valley Steelhead

Surrogate Releases

There have been a total of 5 releases totaling 1,373,848 steelhead in Water Year 2026. JPE for the hatchery releases as of today is 591,419 based on estimated survivals using forecasted water year types (see details in table below). The annual loss threshold, equal to 1% of the JPE, is currently 5,914, but is subject to change with additional steelhead releases.

Table 3: Summary of steelhead hatchery releases in Water Year 2026. JPE calculated using hatchery-specific survival estimates to Delta entry from release location.

Hatchery	Date of Release	Number Released	Estimated Survival	Juvenile Production Estimate
NIM	2025-11-10	233,109	72%	167,838
Coleman	2025-12-15	555,720	38%	211,174
Coleman	2025-12-17	90,019	38%	34,207
FRH	2026-01-06	371,250	36%	133,650
FRH	2026-01-09	123,750	36%	44,550

Table 4: Hatchery-specific survival estimates used for JPE calculations.

Hatchery	Survival Estimate	Source
Coleman NFH	0.205 – 0.433	Sandstrom et al. 2020
Feather River Hatchery	0.09 – 0.45	Kurth 2013
Nimbus Hatchery	0.62 – 0.83	Brodsky et al. 2020

Hatchery	Survival Estimate	Source
Mokelumne River Hatchery	0.25 – 0.33	Del Real et al. 2012

Total loss of hatchery-origin steelhead is 303 or 5.12% of the annual loss threshold. *Note that hatchery origin of salvaged fish cannot be determined at this time and salvage is based on the assumption of similar routing and survival probabilities of individual hatchery releases.*

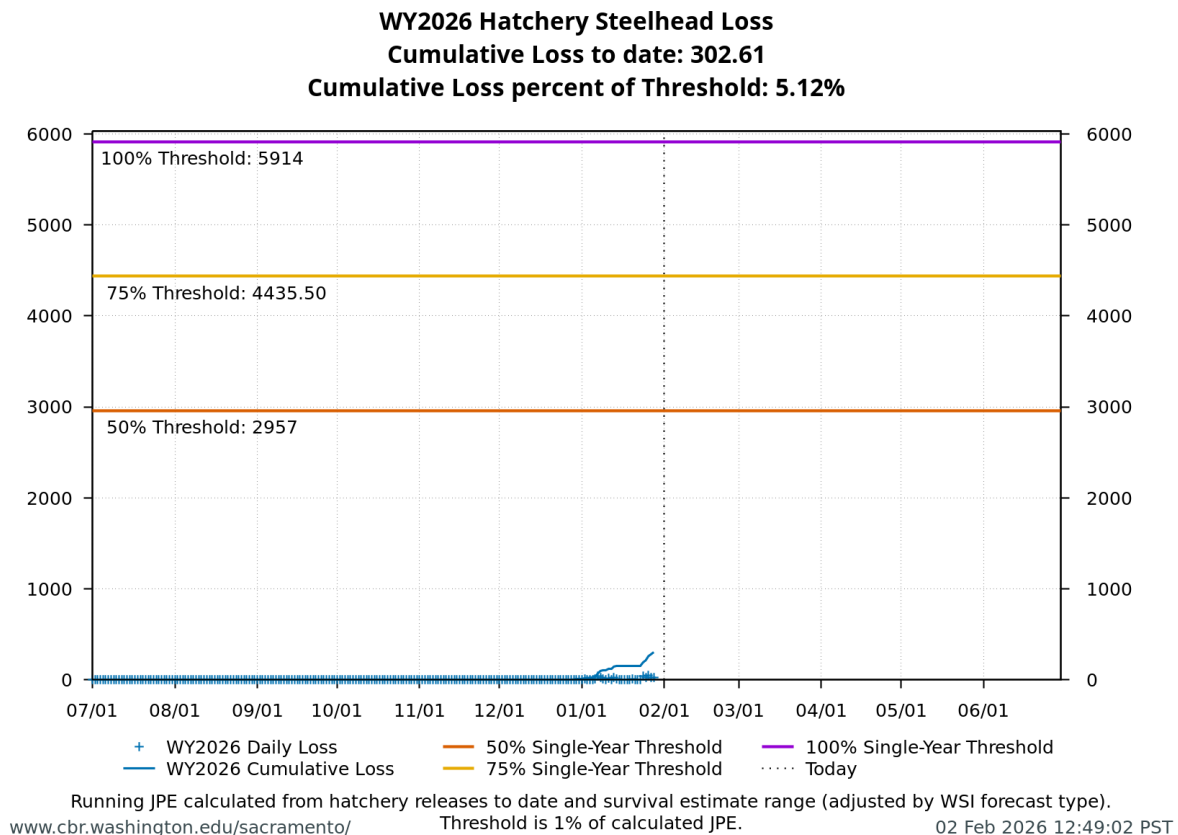


Figure 6: Cumulative loss of hatchery steelhead for WY 2026.

Spring-run Chinook

Current Status

Delta Entry Timing - Historically, as of Feb 01, 3% of LAD spring-run have entered the Delta based on Knights Landing RST catch, 0% have exited the Delta based on Chipps Island Trawl Catch, and 0% have been salvaged.

Table 5: Average percent of annual emigrating population for LAD spring-run Chinook salmon captured at monitoring locations and salvaged at Delta facilities for the past 10 years.

Species	Red Bluff Diversion Dam	Tisdale RST	Knights Landing RST	Sac Trawl (Sher- wood)	Chipps Island Trawl	Salvage
Chinook, LAD Spring- run, Unclipped	15%	17%	28%	3%	0%	0%

Red Bluff Diversion Dam Passage Estimate - As of Jan 28 estimated passage to date of LAD spring-run at Red Bluff Diversion is approximately 0.02 million fish. *Note that outmigration timing overlaps with winter-run and fall-run outmigration, and true spring-run abundance likely differs from these estimates.*

Delta Monitoring - Total catch of LAD spring-run at RSTs at Delta Entry (Tisdale, Knights Landing, Lower Sacramento River) between Jan 19 and Jan 29 is 186 individuals. Total catch at Sacramento Trawl and Beach Seines in the delta between Jan 20 and Jan 30 is 0 individuals. Total catch at Delta Exit at Chipps Island between Jan 20 and Jan 30 is 0 individuals.

Spring-run Surrogate Releases

A total of 805,323 spring-run surrogate fish have been released in Water Year 2026, with an estimated Juvenile Production Estimate (JPE) of 219,852 fish entering the Delta. This includes 805,323 Coleman Late-Fall Run Chinook (JPE: 219,852) released from Coleman National Fish Hatchery across 12 coded-wire tag groups. See details in table below.

Table 6: Spring-run Chinook salmon surrogate releases (all Coleman Late-Fall releases, both production and experimental).

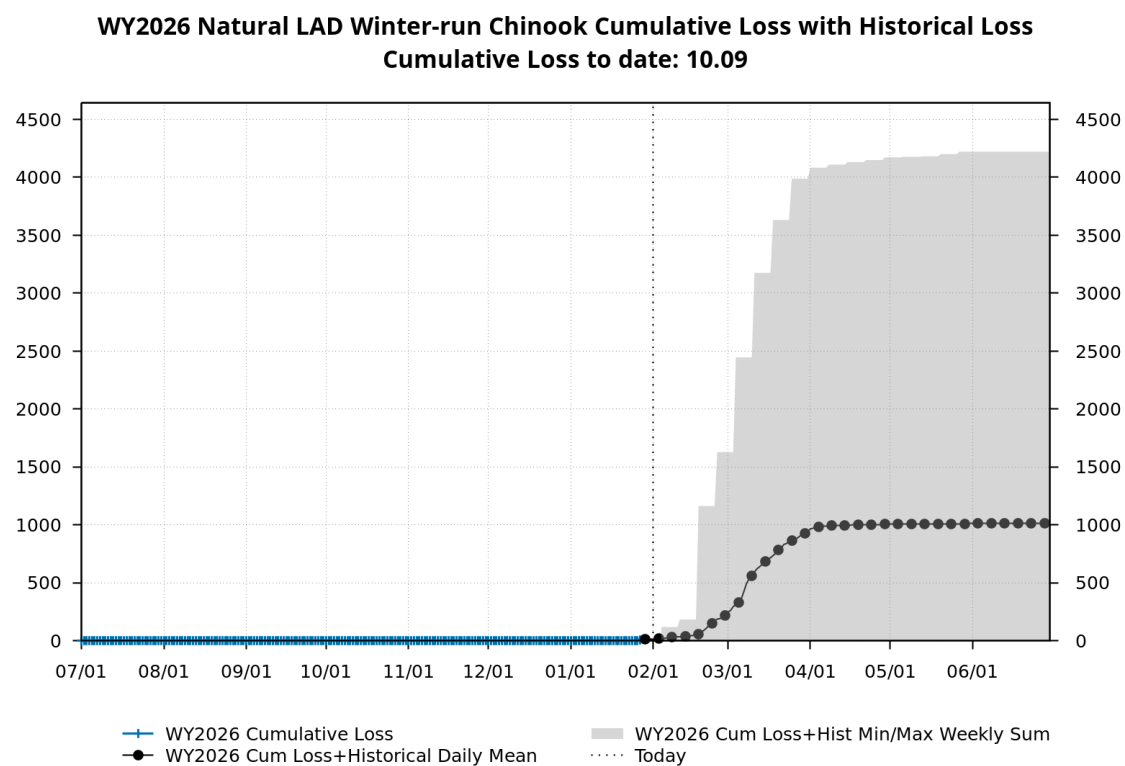
Hatchery	Release Date	Type	# of CWT Fish Released	JPE	ITL (0.5%)	Confirmed CWT Loss Codes
Coleman NFH	2025-11-13	Production	143,346	39,134		8.7 056808, 056809
Coleman NFH	2025-11-17	Experimental	75,119	20,507	376	0.0 056810
Coleman NFH	2025-12-17	Production	468,876	128,002		749.9 053700, 056806, 056811, 056812, 056814, 056815, 056817
Coleman NFH	2025-12-22	Experimental	60,873	16,618	304	257.3 056813
Coleman NFH	2026-01-08	Experimental	67,109	15,591	286	2.9 056816

Annual Loss

The annual loss threshold is 1% of the JPE entering the Delta, which equals 2,199 fish. As of February 01, cumulative loss is 1,019 fish or 46.34% of the annual loss threshold. The single-year incidental take limit (ITL) is 0.5% of the estimated number of each surrogate release group (BiOp Table 184). ITL status by experimental release group: Release Group 1 (2025-11-17): 0 loss of 376 ITL (0%); Release Group 2 (2025-12-22): 257.3 loss of 304 ITL (84.64%); Release Group 3 (2026-01-08): 2.9 loss of 286 ITL (1.01%)

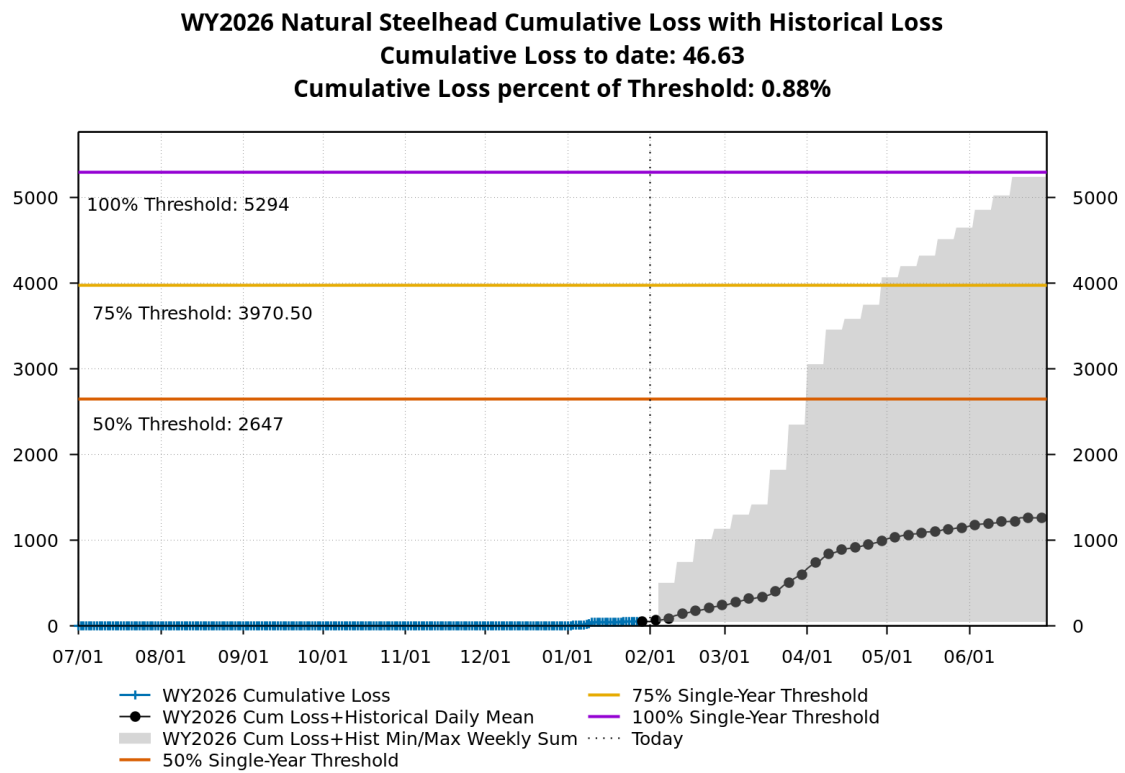
Loss Prediction and Trajectories

The following figures display the cumulative loss trajectories relative to historical years (“Spaghetti Plots”) and the current loss predictor model outputs for winter-run Chinook salmon and steelhead.



Winter-run based on Length-at-Date Delta Model. Historical Loss calculated for WY2009-2018.
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Figure 7: Natural LAD winter-run Chinook salmon loss cumulative to date with historical years and annual loss threshold.



Historical Loss calculated for WY2009-2018.

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Figure 8: Natural Central Valley steelhead loss cumulative to date with historical years and single-year ITL (5,294). There is no annual loss threshold for natural steelhead.

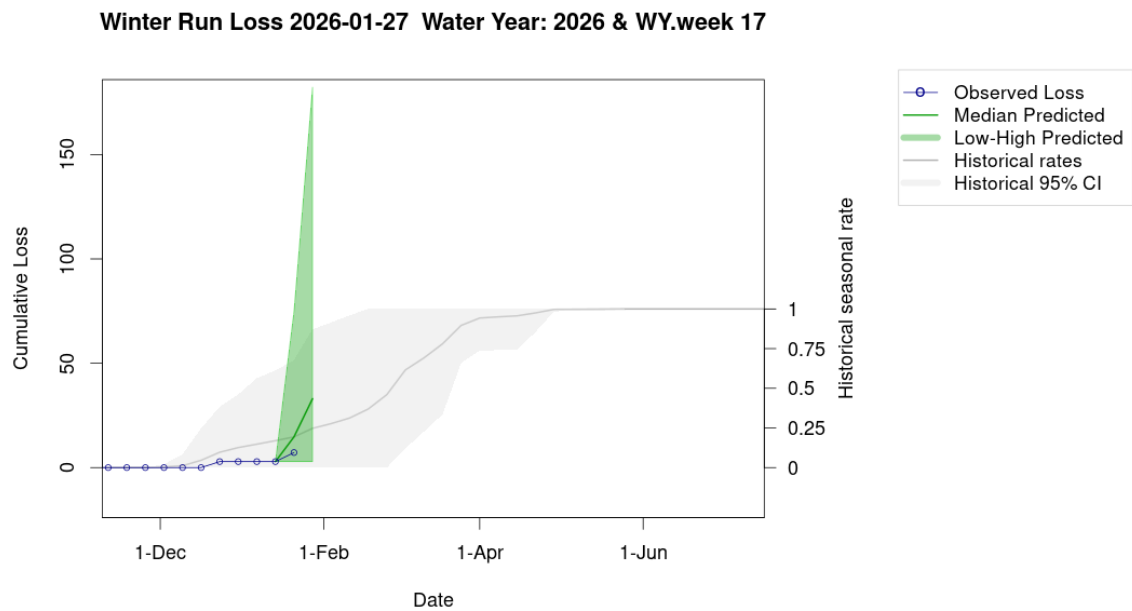


Figure 9: Estimates of winter-run Chinook loss generated by Loss and Salvage Predictor tool.

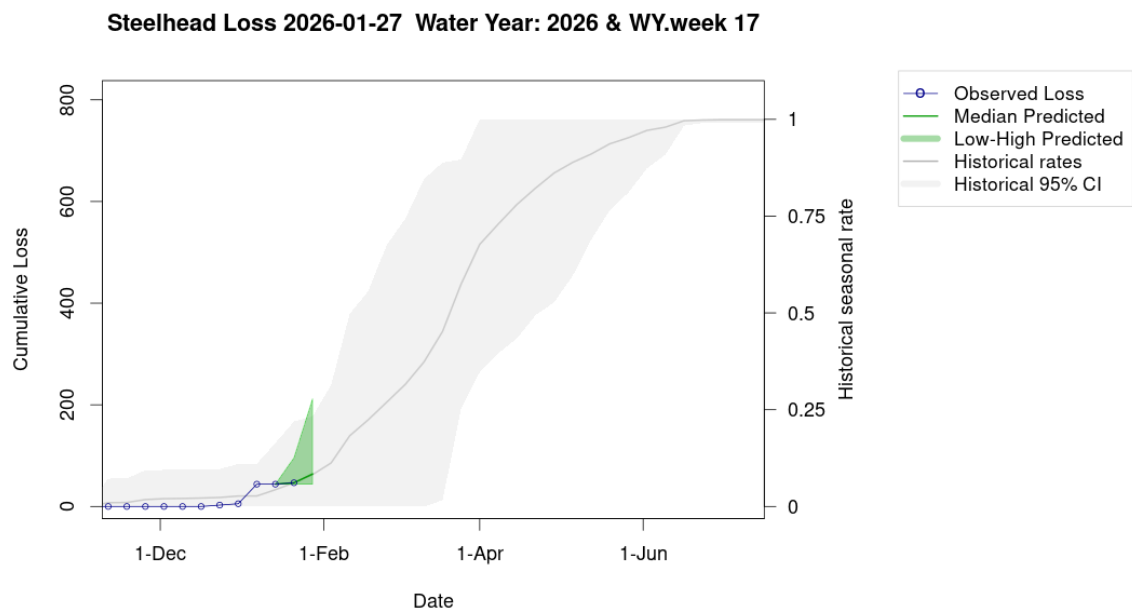


Figure 10: Estimates of steelhead loss generated by Loss and Salvage Predictor tool.

Evaluation

Single-Year Incidental Take Limits (BiOp Table 184): Natural winter-run: 5,922 fish; Hatchery winter-run (Sacramento River): 1,301 fish; Natural steelhead: 5,294 fish; Yearling spring-run surrogates (0.5% of each experimental release group): Release Group 1 (2025-11-17): 0 loss of 376 ITL (0%); Release Group 2 (2025-12-22): 257.3 loss of 304 ITL (84.64%); Release Group 3 (2026-01-08): 2.9 loss of 286 ITL (1.01%)

1. **What is the probability of exceeding natural or hatchery winter-run Chinook salmon loss thresholds or ITLs in the upcoming week?**

LOW RISK: Cumulative loss is currently 0% of the threshold. Current trajectory suggests the threshold is unlikely to be exceeded in the upcoming week.

2. **What is the probability of exceeding spring-run Chinook salmon surrogate yearling loss thresholds or ITLs in the upcoming week?**

LOW RISK: Cumulative loss is currently 46.3% of the threshold. Current trajectory suggests the threshold is unlikely to be exceeded in the upcoming week.

3. **What is the probability of exceeding natural steelhead ITL or hatchery steelhead loss thresholds in the upcoming week?**

INCREASING RISK: While cumulative loss is currently low (5.1%), recent salvage indicates a sharp upward trend.

References

- Brodsky, A., Zeug, S. C., Nelson, J., Hannon, J., Anders, P. J., & Cavallo, B. J. (2020). Does broodstock source affect post-release survival of steelhead? Implications of replacing a non-native hatchery stock for recovery. *Environmental Biology of Fishes*, 103(5), 437–453.
- Del Real, S. C., Workman, M., & Merz, J. (2012). Migration characteristics of hatchery and natural-origin oncorhynchus mykiss from the lower mokelumne river, california. *Environmental Biology of Fishes*, 94(2), 363–375.
- Kurth, R. (2013). *Downstream migration success of feather river fish hatchery steelhead smolts under different release strategies*.
- Sandstrom, P. T., Ammann, A. J., Michel, C., et al. (2020). Low river survival of juvenile steelhead in the sacramento river watershed. *Environmental Biology of Fishes*, 103(5), 531–541.